

Contact Manager Sprint 2

Project Structure

```
ContactManager/
|
├── src/main/java/com/wipro/
|   ├── Contact.java
|   ├── ContactStorage.java
|   ├── HomeServlet.java
|   ├── ContactServlet.java
|   ├── DeleteContactServlet.java
|   ├── EditContactServlet.java
|   └── UpdateContactServlet.java
|
├── src/main/webapp/
|   ├── style.css
|   └── WEB-INF/
|       ├── views/
|       │   ├── index.jsp
|       │   ├── contacts.jsp
|       │   └── edit.jsp
|       └── web.xml
|
```

Application Flow

1. User opens Contact Manager homepage
2. User clicks Add Contact button
3. User enters contact details
4. Contact details stored in ArrayList collection
5. User can view saved contacts
6. User can edit existing contact details
7. Updated contact information displayed in table
8. User can delete contacts from the table

Key Classes

1. Contact.java: Stores contact details such as name, phone, and email.

```
package com.wipro;
public class Contact {
    private String name;
    private String phone;
    private String email;

    public Contact() {
    }
    public Contact(String name, String phone, String email) {
        this.name = name;
        this.phone = phone;
        this.email = email;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getPhone() {
        return phone;
    }

    public void setPhone(String phone) {
        this.phone = phone;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }
}
```

2. ContactStorage.java: Maintains shared ArrayList for contacts.

```
package com.wipro;
import java.util.ArrayList;
import java.util.List;
public class ContactStorage {

    public static List<Contact> contacts = new ArrayList<>();
}
```

3. HomeServlet.java: Loads homepage JSP page.

```
package com.wipro;
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/")
public class HomeServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        request.getRequestDispatcher(
            "/WEB-INF/views/index.jsp")
            .forward(request, response);
    }
}
```

4. ContactServlet.java: Handles adding new contacts.

```
package com.wipro;
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/addContact")
public class ContactServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        String name = request.getParameter("name");
        String phone = request.getParameter("phone");
        String email = request.getParameter("email");

        Contact contact =
            new Contact(name, phone, email);

        ContactStorage.contacts.add(contact);
    }
}
```

```

        request.setAttribute(
            "contactList",
            ContactStorage.contacts);

        request.getRequestDispatcher(
            "/WEB-INF/views/contacts.jsp")
            .forward(request, response);
    }
}

```

5. DeleteContactServlet.java: Deletes contact from ArrayList.

```

package com.wipro;
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/delete")
public class DeleteContactServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        int index = Integer.parseInt(
            request.getParameter("index"));

        ContactStorage.contacts.remove(index);

        request.setAttribute(
            "contactList",
            ContactStorage.contacts);

        request.getRequestDispatcher(
            "/WEB-INF/views/contacts.jsp")
            .forward(request, response);
    }
}

```

6. EditContactServlet.java: Loads selected contact details into edit form.

```

package com.wipro;
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;

```

```

import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/edit")
public class EditContactServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        int index = Integer.parseInt(
            request.getParameter("index"));

        Contact contact =
            ContactStorage.contacts.get(index);

        request.setAttribute("contact", contact);
        request.setAttribute("index", index);

        request.getRequestDispatcher(
            "/WEB-INF/views/edit.jsp")
            .forward(request, response);
    }
}

```

7. UpdateContactServlet.java: Updates edited contact details.

```

package com.wipro;
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/update")
public class UpdateContactServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        int index = Integer.parseInt(
            request.getParameter("index"));

        String name = request.getParameter("name");
        String phone = request.getParameter("phone");
        String email = request.getParameter("email");
    }
}

```

```

        Contact updatedContact =
            new Contact(name, phone, email);

        ContactStorage.contacts.set(
            index,
            updatedContact);

        request.setAttribute(
            "contactList",
            ContactStorage.contacts);

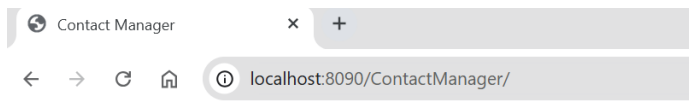
        request.getRequestDispatcher(
            "/WEB-INF/views/contacts.jsp")
            .forward(request, response);
    }
}

```

JSP Pages

- 8. index.jsp: Displays homepage and contact form.
- 9. contacts.jsp: Displays saved contacts with Edit and Delete buttons.
- 10. edit.jsp: Displays editable form for updating contact details.
- 11. style.css: Contains responsive styling for forms, buttons, and tables.
- 12. web.xml: Defines application configuration and welcome file.

Output



Contact Manager

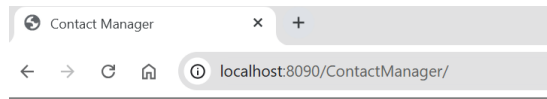
Organize and manage your contacts efficiently

[+ Add New Contact](#)

Saved Contacts

Name Phone Email Edit Delete

Adding a contact:



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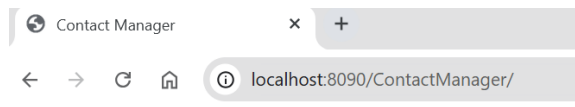
[+ Add New Contact](#)

Add Contact

Full Name
Phone Number
Email Address

Saved Contacts

Name Phone Email Edit Delete



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[+ Add New Contact](#)

Saved Contacts

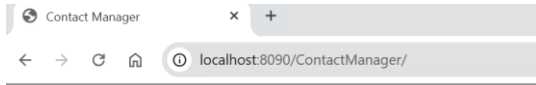
Name	Phone	Email	Edit	Delete
Aishwarya	998877665	aishu@gmail.com	Edit	Delete

Editing a contact:



Edit Contact

Name
Phone
Email



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Saved Contacts

Name	Phone	Email	Edit	Delete
Aishwarya	998877665	aishwarya@gmail.com	Edit	Delete

Adding another contact:

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Add Contact

Full Name
Phone Number
Email Address
[Save Contact](#)

Saved Contacts

Name	Phone	Email	Edit	Delete
Aishwarya	998877665	aishwarya@gmail.com	Edit	Delete

Deleting a contact:

Contact Manager

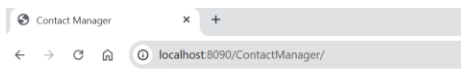
Organize and manage your contacts efficiently

Contact Added Successfully

[+ Add New Contact](#)

Saved Contacts

Name	Phone	Email	Edit	Delete
Aishwarya	998877665	aishwarya@gmail.com	Edit	Delete
abcde	4455667788	abcde@gmail.com	Edit	Delete



Contact Manager

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[+ Add New Contact](#)

Saved Contacts

Name	Phone	Email	Edit	Delete
Aishwarya	998877665	aishwarya@gmail.com	Edit	Delete